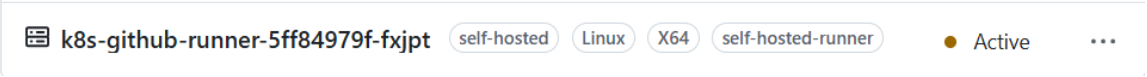
Set up a Kubernetes pod as Self-hosted runner for GitHub Actions pipe line



Step1: Clone the Repository to VS Code

Step2: Update the GitHub Personal Access Token

Step3: Update Base64 Encoded Values in `secret.yaml`

- Use the following command to encode your token or other sensitive data:

```
echo -n "ghp_your_generated_token" | base64
```

- Open the `secret.yaml` file in your preferred text editor.
- ocate the `data` section and Replace the old base64 value with new one generated in step1

```
  data:
    TOKEN: "your-new-base64-encoded-value"
```

Step4: Update GitHub Username and Repository Name Secret in `ge-deployment.yaml`

- Open the `ge-deployment.yaml` file in your preferred text editor.
- Locate the section where the GitHub username and repository name secrets are defined.
- Update the secret for the GitHub username:

```
  - name: GITHUB_USERNAME
    value: "your-new-username"
```

- Update the secret for the repository name:

```
  - name: REPOSITORY_NAME
    value: "your-new-repository-name"
```

step5: Deploy the Self-Hosted Runner Using `kubectl`

- Ensure you have `kubectl` installed and configured to interact with your Kubernetes cluster.
- Navigate to the directory containing your Kubernetes manifests:

```
cd /home/setup-k8s-runner
```

- Apply the manifests to deploy the runner:

```
kubectl apply -f selfthosted-runner
```

- Verify that the pods are running:

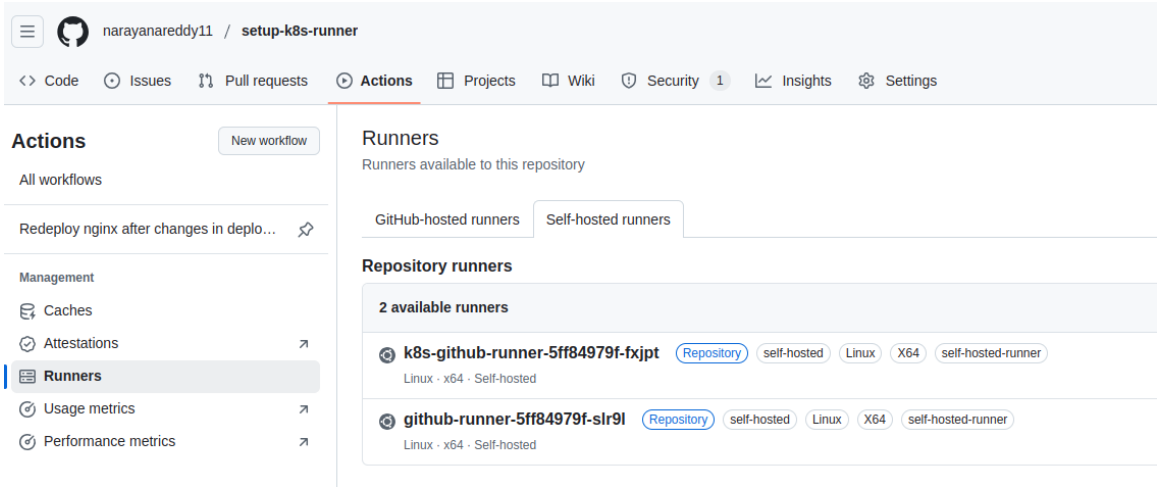
```
kubectl get pods
```

- Check the logs of the runner pod to ensure it is registered successfully:

```
kubectl logs <runner-pod-name>
```

step6: Verify the Runner Registration in Github

- After applying the changes, navigate to your GitHub repository.
- Go to **Settings> Actions > **Runners.**
- Verify that the Kubernetes runner is listed as an active runner.



Step7. Configure the Runner to Monitor Changes in the `nginx` Folder

- Ensure your GitHub Actions workflow is set up to trigger on changes to the `nginx` folder. Update your `.github/workflows/<workflow-file>.yml` file with the following:

```
on:
  push:
    paths:
      - 'nginx/**'
```

- Verify that the runner picks up and executes jobs when changes are made to the `nginx` folder.